

Question	Answer	Marks
3(a)	$E_{(K)} = \frac{1}{2}mv^2$ $110 = \frac{1}{2} \times 5.5 \times v^2$	C1
	$v = 6.3 \text{ m s}^{-1}$	A1
3(b)	$(\Delta)E_{(P)} / 20 = mg(\Delta)h$ OR $(\Delta)E_{(P)} / 20 = mgx_0$	C1
	$(x_0 =) 20 / 5.5 \times 9.81 = 0.37 \text{ (m)}$ Allow $(\Delta)h$ for x_0	A1
3(c)	$[\max (E_{(P)})] = 110 + 20 = 130 \text{ J}$	A1

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3(d)	$(\max) E_{(P)} / 130 = \frac{1}{2} F_{(0)} x_{(0)}$	C1
	$(F_0 =) 2 \times 130 / 0.37 = 700 \text{ (N)}$	A1
	or $(\max) E_{(P)} / 130 = 1 / 2 k x_{(0)}^2$ $(F_0 =) k x_{(0)}$	(C1)
	$(F_0 =) 2 \times 130 \times 0.37 / (0.37)^2 = 700 \text{ (N)}$	(A1)
3(e)(i)	$(\text{weight}) = 5.5 \times 9.81$	C1
	resultant force = $700 - (5.5 \times 9.81)$ = 650 N	A1
3(e)(ii)	$F = ma$	C1
	$a = 650 / 5.5$ or $(700 / 5.5) - 9.81$ $a = 120 \text{ m s}^{-2}$	A1

9702/24

Cambridge International AS & A Level – Mark Scheme

May/June 2025

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1(c)(i)	(copper that) transfers / converts energy	D1